Hydraulic Turbines

Dr. Alireza Riasi

Second Project

Design of a Horizontal Axis Marine Current Turbine

Students

Hamid Motallebi

Hossein Arabzadeh

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School of Mechanical Engineering

College of Engineering, University of Tehran

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Abstract

In this project, the aerodynamic and hydrodynamic design of a Horizontal-Axis Marine Current Turbine (HAMCT) was carried out with the objective of extracting 200 kW of power at a reference flow velocity of 1.5 m/s. To achieve the optimal blade geometry, the Blade Element Momentum Theory (BEMT) was employed, and the iterative calculations were developed within the MATLAB software environment. The turbine blade was divided into 9 radial elements, utilizing the NACA 44xx airfoil family with a variable thickness distribution, ranging from 41% at the root to 16% at the tip. The blade twist angle distribution was optimized via parabolic curve fitting based on the operational performance criteria. The analytical results demonstrate that the designed rotor, with a diameter of 20.76 m and a rotational speed of 6.89 rpm, achieves a favorable power coefficient of $C_p = 0.4144$ and delivers a mechanical power of 236.8 kW. This provides an adequate safety margin to compensate for the mechanical and electrical losses within the overall system. Finally, the three-dimensional coordinates of the blade were extracted for transfer to CAD software and subsequent structural analyses.

Keywords: Horizontal-Axis Marine Current Turbine (HAMCT), Blade Element Momentum Theory (BEMT), NACA 44xx airfoil, variable thickness distribution, twist angle optimization, power coefficient (C_p), rotor design, MATLAB, QBlade, hydrodynamic performance.

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Introduction

The development of energy extraction methods from renewable sources represents one of the most significant engineering challenges in recent decades. Among these resources, tidal and ocean current energy holds a distinctive position due to its high predictability and the considerable energy density of water compared to air. Horizontal-Axis Marine Current Turbines (HAMCTs) are among the most prevalent devices utilized for harnessing this type of energy.

The objective of this project is to carry out the hydrodynamic and geometric design of a rotor for a horizontal-axis marine current turbine. The reference design conditions are defined based on a constant water flow velocity of 1.5 m/s and a target power output of 200 kW. For the turbine blades, the asymmetric NACA 44xx airfoil family is employed. This particular series of airfoils, owing to their suitable thickness at the trailing edge, not only delays the onset of the stall phenomenon but also exhibits excellent structural resistance against bending stresses induced by hydrodynamic water loads, thereby reducing the likelihood of fatigue failure.



Figure 1: Schematic view of a Horizontal-Axis Marine Current Turbine (HAMCT) in operation underwater[1].

The primary design parameters and initial assumptions of the problem, which serve as the basis for the analytical computations (BEM) and subsequent coding, are presented in the tables below:

Table 1:Main Design Specifications

Parameter	Symbol	Value	Unit
Free-stream flow velocity	C_{x1}	1.5	m/s
Target output power	P	200	kW
Tip Speed Ratio (TSR)	J	5	-
Target power coefficient	C_p	0.35	-

Blade section airfoil type	—	NACA 44xx	-

Table 2: Initial Assumptions of the Problem

Parameter	Symbol	Value	Unit
Sea water density	ρ	1000	kg/m³
Number of blades	Z	3	—
Hub-to-tip diameter ratio	D_{hub}/D_{tip}	0.1	—
Design angle of attack	α_D	9	degrees (°)
Number of blade elements for analysis	N	9	—

Methodology

2-1 Blade Element Momentum Theory

For the design and hydrodynamic performance analysis of the turbine blade, the Blade Element Momentum Theory (BEMT) was employed. This method is essentially a combination of the Momentum Theory applied to an annular streamtube (control volume) and the Blade Element Theory, which is used to calculate Lift and Drag forces on different sections along the blade. Within this theoretical framework, two key simplifying assumptions are made:

- There is no hydrodynamic interaction between different blade sections (elements).
- The forces acting on the blade are solely derived from the lift and drag coefficients of the respective sections.

Since the governing equations of this method are mutually interdependent (coupled), their solution necessitates an iterative (trial-and-error) approach.

2-2 Governing Equations of the Design

By discretizing the blade into N radial elements of length Δr, the calculations are carried out at the mean radius of each element, denoted as r_m. The flow angle (φ) and the angle of attack (α) are obtained from the following relations[3]:

[2.1] $$tan\,\varphi = \frac{R_t}{Jr_m}\left(\frac{1-a}{1+a'}\right)$$

[2.2] $$\alpha = \varphi - \beta$$

Where a and a' are the axial and tangential induction factors, respectively, and β is the blade twist angle

The chord length (l) for each element is calculated based on the design angle of attack:

[2.3] $$l = \frac{1}{Z} \times \frac{16\pi r_m}{C_L} \times \sin^2\left(\frac{1}{3}\arctan\left(\frac{R_t}{Jr_m}\right)\right)$$

In order to account for the losses induced by tip vortices, the Prandtl tip-loss correction factor (F) is employed as follows:

[2.4] $$d = R_t - r_m$$

[2.5] $$s = \frac{\pi d}{\frac{1}{2}Z\left(1-\frac{r_m}{R_t}\right)\sqrt{1+J^2}}$$

[2.6] $$F = \frac{2}{\pi} arccos\left(exp\left(\frac{-\pi d}{s}\right)\right)$$

By defining the auxiliary parameters $\lambda = \frac{ZlC_L}{8\pi r_m}$ and $\epsilon = \frac{C_D}{C_L}$, the updated values of the induction factors for the subsequent iteration step are obtained from the following formulas:

[2.7] $$a = \frac{1}{1 + \frac{F}{\lambda}\frac{\sin^2\varphi}{\cos\varphi + \epsilon\sin\varphi}}$$

[2.8] $$a' = \frac{1}{\frac{F}{\lambda}\frac{\sin\varphi\cos\varphi}{\sin\varphi - \epsilon\cos\varphi} - 1}$$

This iterative process is continued until the difference between the values of a and a' in two consecutive iterations falls below the predefined convergence tolerance (allowable error).

2-3 Extraction of the Twist Angle Distribution

The blade twist angle (β) must be distributed such that the airfoil sections along the blade span operate within a range that yields approximately 80% of their maximum lift force. Due to the variation in linear velocity from the root to the tip of the blade, the flow angle continuously changes along the radius.

To achieve a continuous and manufacturable geometry, the flow angles were calculated at three key sections (root, mid-span, and tip) based on the tip speed ratio $J = 5$. Subsequently, considering the optimal angles of attack, the twist angle was extracted, and finally, a parabolic equation (second-order polynomial) was fitted to these discrete points (curve fitting). This approach ensures a smooth and continuous distribution of the twist angle β along the blade span. The specific parabolic equation derived for this rotor, with the twist angle expressed in degrees, is as follows:

[2.9] $$\beta = 78.31\left(\frac{r_m}{R_t}\right)^2 - 131.44\left(\frac{r_m}{R_t}\right) + 61.08$$

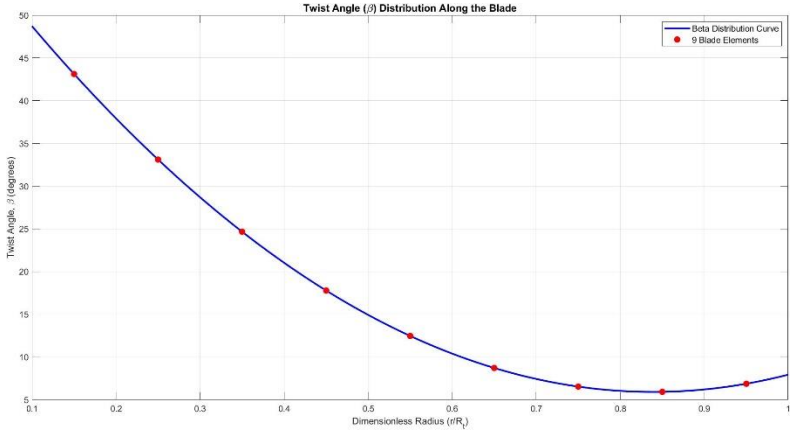


Figure 2: Plot of the twist angle (β) distribution versus the dimensionless radius.

2-4 Calculation of the Final Rotor Performance

Following the complete convergence of the induction factors across all sections, the total axial force (X) and the total torque (T) of the rotor are obtained through numerical integration (i.e., summation) of the elemental values:

[2.10] $$X = \frac{1}{2}\rho Z R_t C_{x1}^2 \sum\left(\frac{1-a}{\sin\varphi}\right)^2 C_L\cos\varphi\frac{\Delta r}{R_t}$$

[2.11] $$T = \frac{1}{2}\rho Z \Omega^2 R_t^4 \sum \left(\frac{1+a'}{\cos\varphi}\right)^2 \left(\frac{r_m}{R_t}\right)^3 C_L \sin\varphi \frac{\Delta r}{R_t}$$

Finally, the output power (P) and the power coefficient (C_p) of the rotor are calculated as follows:

[2.12] $$P = T\Omega$$

[2.13] $$C_p = \frac{P}{\frac{1}{2}\rho A C_{x1}^3}$$

Design Calculations and Results

3-1 Preliminary Calculations and Geometric Sizing

Prior to entering the BEMT iterative loop, it is necessary to calculate the reference rotor dimensions and rotational speeds based on the design objectives. With the water flow velocity ($C_{x1} = 1.5$ m/s), the target output power ($P = 200$ kW), and an assumed power coefficient ($C_p = 0.35$), the swept area of the rotor (A) and the blade tip radius (R_t) are derived as follows[2]:

$$A = \frac{P}{\frac{1}{2}\rho C_{x1}^3 C_p} = 338.624\ m^2 \Longrightarrow R_t = 10.382\ m \quad [3.1]$$

Given the tip speed ratio ($J = 5$), the angular velocity of the rotor (Ω) is obtained as follows:

$$\Omega = \frac{J \cdot C_{x1}}{R_t} = 0.7224\ rad/s \Longrightarrow N = 6.898\ rpm \quad [3.2]$$

Additionally, the hub radius was taken as $R_{hub} = 1.038$ m, based on the assumption that it constitutes 10% of the total blade radius.

3-2 Blade Element Specifications and Reynolds Number

The blade span was discretized into 9 equal elements ($\Delta r = 1.0382$ m). The thickness ratio of the NACA 44xx airfoil family decreases from 41% at the root to 16% at the tip, so as to maximize the aerodynamic efficiency at the blade tip while maintaining adequate structural strength at the root. The operating Reynolds number (Re) for each section was computed based on the relative flow velocity and the local chord length:

Table 3: Specifications of the Blade Elements and Selected Airfoils

Element No.	Radius Ratio (r_m/R_t)	Airfoil Type	Reynolds Number (Re)
1	0.15	NACA 4441	3,607,623
2	0.25	NACA 4438	4,146,873
3	0.35	NACA 4434	4,345,316
4	0.45	NACA 4431	4,458,473
5	0.55	NACA 4427	4,508,892
6	0.65	NACA 4423	4,526,471
7	0.75	NACA 4418	4,585,570

Element No.	Radius Ratio (r_m/R_t)	Airfoil Type	Reynolds Number (Re)
8	0.85	NACA 4416	4,569,721
9	0.95	NACA 4416	4,575,953

3-3 Results of the Aerodynamic Analysis of the Sections

By implementing the iterative trial-and-error algorithm in the MATLAB software environment and applying a relaxation factor to prevent divergence of the governing equations, the axial (a) and tangential (a') induction factors were converged for all nine elements. The following table presents the converged values along with the geometric distribution of the chord length and twist angle:

Table 4: Convergence Results of the Blade Element Theory for the Rotor Sections

Element No.	Section Radius r_m (m)	Factor a	Factor a'	Flow Angle φ (deg)	Twist Angle β (deg)	Angle of Attack α (deg)	Chord Length l (m)
1	1.56	0.1092	0.1626	45.63	43.13	2.51	1.93
2	2.60	0.0928	0.0537	34.59	33.12	1.47	1.73
3	3.63	0.1030	0.0302	26.48	24.67	1.81	1.44
4	4.67	0.1313	0.0225	20.71	17.79	2.92	1.21
5	5.71	0.1653	0.0181	16.62	12.48	4.14	1.03
6	6.75	0.1958	0.0147	13.72	8.73	4.99	0.89
7	7.79	0.2166	0.0118	11.68	6.55	5.13	0.79

Element No.	Section Radius r_m (m)	Factor a	Factor a'	Flow Angle φ (deg)	Twist Angle β (deg)	Angle of Attack α (deg)	Chord Length l (m)
8	8.82	0.2278	0.0095	10.22	5.93	4.28	0.70
9	9.86	0.2592	0.0081	8.80	6.89	1.92	0.63

The distribution of the axial induction factors remains below the critical threshold of 0.5 across all sections, indicating stable flow conditions and the absence of entry into the Turbulent Wake State. Furthermore, the decreasing trend of the chord length (from 1.93 m at the root to 0.63 m at the tip), in addition to improving the aerodynamic conditions, ensures the mechanical strength of the blade.

3-4 Final Rotor Performance

Through numerical integration of the local sectional values, the overall performance of the turbomachine was derived. The final values indicate that the initial design targets have been met and even exceeded:

Table 5: Overall Performance of the Designed Turbine

Performance Parameter	Value	Unit
Total Axial Force (Thrust)	199.35	kN
Total Rotor Torque	327.80	kN·m
Mechanical Output Power	236.80	kW
Rotor Power Coefficient (C_p)	0.4144	—

The extracted hydrodynamic power (236.8 kW) provides a substantial safety margin over the target power (200 kW). This safety margin, under real operating conditions of the turbine, compensates for the losses attributable to the mechanical efficiency of the gearbox and the electrical efficiency of the generator, thereby enabling the delivery of the rated power of 200 kW to the grid. The power coefficient of 0.4144 is also considered an optimal and highly favorable efficiency for horizontal-axis turbines at this scale.

Three-Dimensional Modeling and Software-Based Validation

4-1 Geometry Creation in SolidWorks

Following the extraction of the twist angle and chord length distributions from the MATLAB code, the three-dimensional coordinates of the airfoil sections were generated for the nine blade elements. These sections were then imported into the SolidWorks environment as three-dimensional curves. Using the Lofted Boss/Base command, a continuous surface was passed through these sections to form the solid volume of the blade. This precise and seamless modeling approach enables the geometry to be exported for future structural and finite element analyses (e.g., stress and vibration investigations).

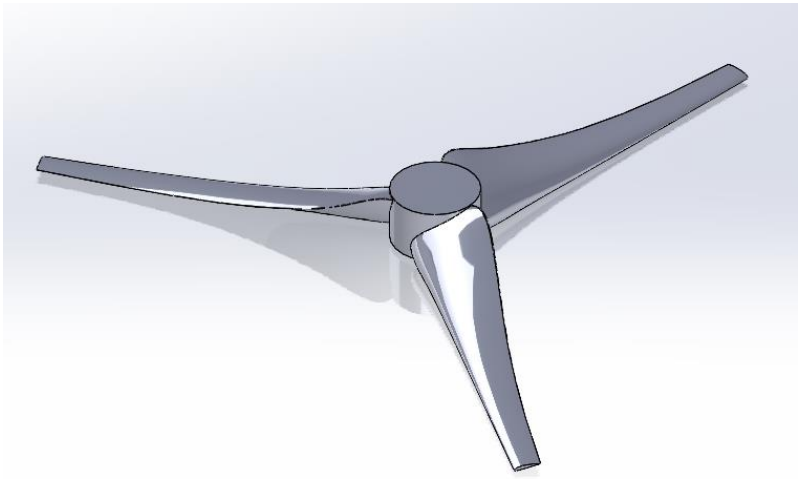


Figure 3: Front view of the HAMCT rotor assembly

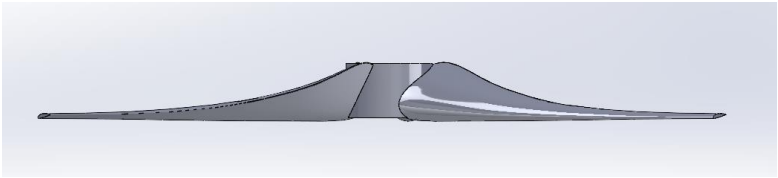


Figure 4: Side view of the rotor and hub assembly



Figure 5: Three-dimensional isometric view of a single extracted blade

4-2 Aerodynamic Simulation in QBlade Software

For the purpose of validating the calculations performed using the Blade Element Momentum (BEM) theory, the designed geometry was imported into the QBlade software environment. In this software, following the generation of 360-degree polar curves for the NACA 44xx sections at their corresponding Reynolds numbers, a simulation of the rotor performance was carried out.

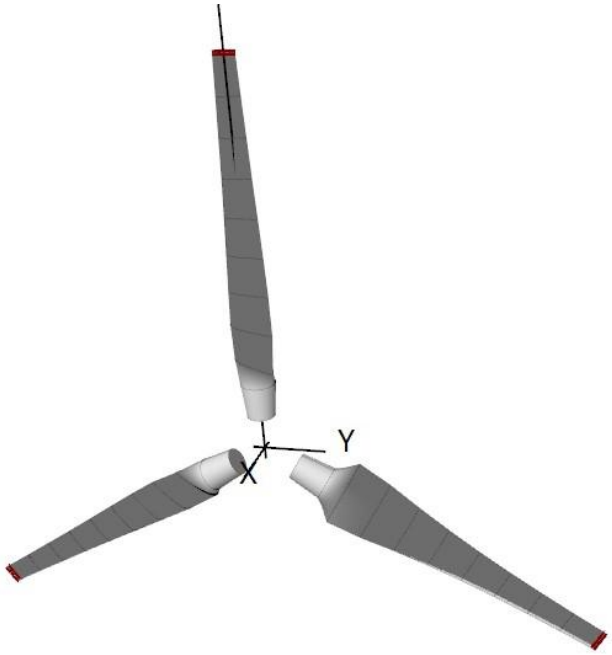


Figure 6: Three-dimensional isometric view of the complete three-bladed rotor assembly in the QBlade environment.

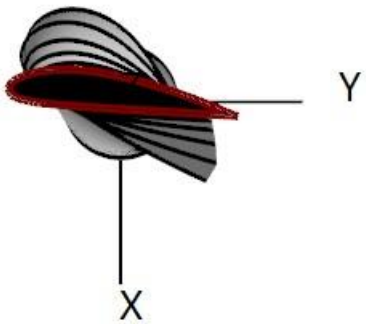


Figure 8: Isometric view of a single blade extracted from the QBlade rotor model.



Figure 7: Span-wise twist angle variation illustrated on the blade surface.

4-3 Analysis of QBlade Performance Curves

The simulation outputs from the QBlade software effectively confirm the physical and aerodynamic behavior of the rotor. In the following, two key parameters from these outputs are examined:

(a) Power coefficient versus tip speed ratio (C_p vs. TSR)

The figure below illustrates the variation of the rotor power coefficient against different rotational speeds.

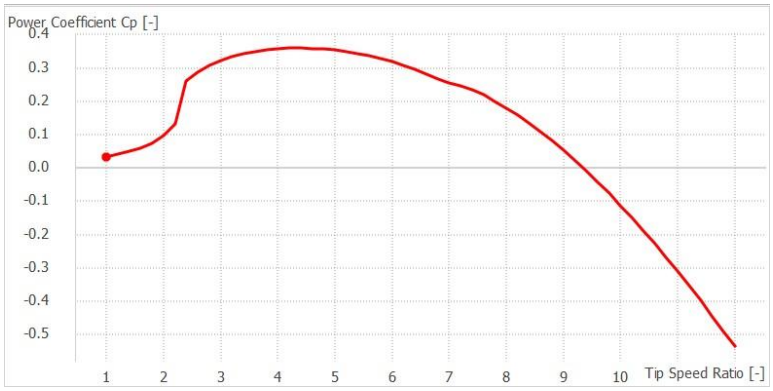


Figure 9: Variation of the rotor power coefficient (C_p) versus the tip speed ratio (TSR), obtained from the aerodynamic simulation in QBlade.

As observed in the above figure, the power coefficient curve starts from low values at low tip speed ratios and rises with a steep slope. This curve reaches its peak (the highest hydrodynamic efficiency) within the TSR range of 4 to 5, recording a value exceeding 0.35. This behavior is in complete agreement with the initial assumptions of the problem, which set the optimal operating point (target TSR) at 5. The sharp drop in the curve at TSRs above 7 is also attributed to the onset of stall and the increased drag forces at high rotational speeds.

(b) Distribution of the axial induction factor along the blade span

The distribution of the axial induction factor (a) indicates the magnitude of the reduction in water flow velocity as it passes through the rotor disk and serves as a criterion for assessing the stability of the flow.

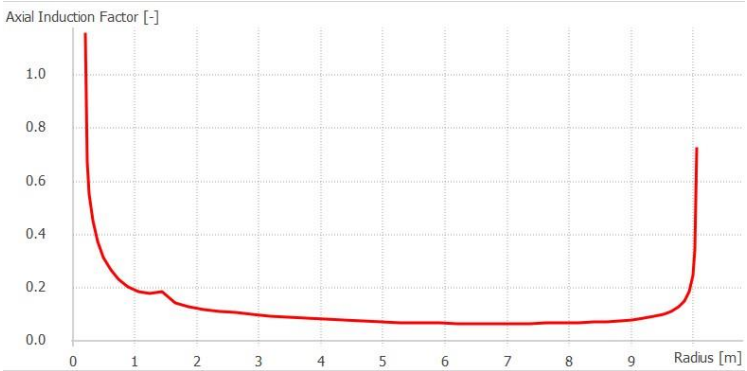


Figure 10: Distribution of the axial induction factor (a) along the blade span, illustrating the effects of root vortices and the tip loss phenomenon.

Based on the above figure, the flow behavior along the blade span (from 0 to approximately 10.38 m) can be divided into three distinct physical regions:

1. **Root region (near 0 m):** The induction factor is considerably high due to the very low linear velocity of the sections close to the hub, as well as the effects of root vortices.
2. **Mid-span region (between 2 and 9 m):** Along the effective aerodynamic portion of the blade, the axial induction factor reaches a stable and highly favorable value (ranging between 0.1 and 0.2). Remaining well below the critical threshold of 0.5, this indicates fully stable flow conditions and confirms that the blade does not enter the Turbulent Wake State.
3. **Tip region (beyond 9.5 m):** Near the blade tip, the induction factor experiences a sharp increase once again. This phenomenon is entirely natural and reflects the physical effect of **tip loss**, which arises from the leakage of fluid from the high-pressure side to the low-pressure side of the airfoil, leading to the formation of tip vortices.

Conclusion

In this project, the complete hydrodynamic and geometric design procedure of a Horizontal-Axis Marine Current Turbine (HAMCT) was presented with the objective of extracting a rated power of 200 kW. Based on the obtained results, the application of the Blade Element Momentum Theory (BEMT) in conjunction with iterative numerical algorithms proves to be an efficient and reliable method for deriving the optimal rotor geometry. By selecting the NACA 44xx airfoil series, employing a thickness distribution decreasing from 41% at the root to 16% at the tip, and extracting a precise parabolic equation for the twist angle distribution, the mechanical output power reached 236.8 kW. This value not only satisfies the target power of 200 kW but also provides an adequate safety margin to compensate for mechanical and electrical losses within the drivetrain system. Furthermore, achieving a power coefficient of 0.4144 confirms the favorable aerodynamic performance of the rotor at the design operating point.

The analysis of the results indicates that the derived blade geometry establishes a proper engineering balance between fluid dynamic efficiency and structural stiffness requirements. Maintaining the axial induction factor within the allowable range (below 0.5) ensures flow stability and prevents the blade from entering the Turbulent Wake State. Validation of these data through simulations carried out in the QBlade software further corroborated the accuracy of the analytical computational procedure.

Among the limitations of the present design phase are the assumptions of steady-state and uniform flow in the modeling. The complex three-dimensional flow effects, wake interactions with the supporting structure, the detrimental phenomenon of cavitation, and fluctuating loads induced by surface gravity waves have not been taken into account in these calculations.

For the further development of this project and to bring the design closer to real-world operational conditions, it is recommended that future studies incorporate Fluid-Structure Interaction (FSI) analyses. Given the oscillatory nature of hydrodynamic loads on rotating equipment, transferring the CAD model to finite element software is essential for detailed investigations of structural vibrations, harmonic analysis, and fatigue life prediction of the blades. Additionally, the design of an active control system (such as a pitch control mechanism) to adjust the blade angles under varying flow velocities could significantly enhance the dynamic stability of the system.

Overall, the integration of analytical formulations, numerical computations, and modeling within this project has established a standardized framework for the practical design of marine rotary equipment. The outcomes of this work may serve as a basis for the engineering design phase in future projects aimed at the construction and development of ocean energy extraction systems.

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